INTERNATIONAL MONETARY FUND

CHAD

STAFF REPORT FOR THE 2012 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Chad's public domestic and external debt indicators have worsened since 2010 and the large amount of non-concessional borrowing contracted in 2011 entails a high risk of debt distress. The assumptions underpinning the debt sustainability analysis (DSA) have been revised from last year to reflect significant increases in projected oil production, prices, and related government revenues, and higher external borrowing. The latter reflects mainly the expected implementation of the $2 billion non-concessional Master Facility Agreement (MFA) signed with the Eximbank of China in August 2011. The updated baseline scenario shows that, if the MFA is implemented in its current form, all prudential thresholds will be breached within the projection horizon. An alternative scenario shows that all prudential thresholds would be observed if: (i) the MFA were concessional (with a grant element of 35 percent); (ii) it was implemented over ten years (instead of five); and (iii) other external borrowing was drawn mainly from concessional sources and its amount remained moderate.

BACKGROUND

Recent Developments in Public External Debt

1. **Chad's external public debt burden indicators improved considerably in 2001–09, thanks to the strong oil-driven growth and low borrowing from abroad**. Nonetheless, the drop in international oil prices in 2009 pushed the external public debt-to-GDP ratio from about 20 percent in 2008 to 23 percent in 2009, revealing Chad's vulnerability to a negative oil price shock.

2. **In 2010–11, the authorities borrowed abroad on non-concessional terms and external (public and publicly guaranteed) debt increased**.[1] As a result, the rate of debt accumulation (the year-to-year change in present value (PV) of debt relative to previous year's GDP) spiked between 2009 and 2010–11, in part due to the low grant element of new borrowing (both from Libya and China). Because of a higher-than-anticipated recourse to external borrowing (including guaranteed debt) and despite a better-than-anticipated overall fiscal situation (a surplus of 1.6 percent of non-oil GDP), helped by the record-high oil revenues, the external public debt-to-GDP ratio exceeded 26 percent at end-2011 (compared to 24 percent anticipated in the 2011 DSA).

3. **In August 2011, the authorities signed a Master Facility Agreement with the Eximbank of China (MFA)**. The MFA is an umbrella contract for individual facilities that would be activated to finance eligible projects for a total amount of up to $2 billion (or 18 percent of the 2012 GDP) over a five-year period. Projects to be funded under the MFA include: (i) a new international airport for N'Djamena; (ii) a gas power plant (100 MW); (iii) construction of 400–500 km of roads; (iv) a two-lane road bridge near N'Djamena; (v) cultivation of 20,000 ha of agricultural land; and (vi) a cement plant (with a capacity of 600,000 tons per year). The financial terms of the MFA are non-concessional. The borrowing terms include: (i) an interest rate of 1.5 percent; (ii) a risk fee of 4.0 percent (a top up of the interest rate); (iii) a management fee (upfront) of 0.6 percent on each individual facility; (iv) a commitment fee of 0.4 percent levied on the unutilized portion of each individual facility; (v) the limit on the tenor of each facility not to exceed 20 years; and (vi) a grace period of five years. The payment obligations under the MFA are collateralized on oil export receipts. Thus, the signing of the MFA is significantly affecting the baseline scenario which was retained by staffs in the 2011 DSA.

[1] Among others, the authorities took a $300 million loan from the Libyan Foreign Bank, which was disbursed over 2010-11 and guaranteed about $600 million of debts disbursed in 2010–11, comprising the loan to finance the 40 percent share by the state oil company (SHT) in the oil refinery built by the China National Petroleum Corporation (CNPC), and a loan to finance state shares in the Baoré cement factory.

Text Table 1. Chad: External Debt Stock at Year-End, 2001–2011
(Billions of CFA francs)

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011 Est.
Total	794.7	786.6	736.9	797.2	898.9	896.2	794.0	782.3	781.8	1,066.8	1317.1
(percent of GDP)	63.4	56.8	46.3	34.2	29.0	27.2	23.6	20.9	23.4	25.2	26.4
Multilateral	678.1	687.7	652.5	715.3	810.2	805.5	718.6	706.6	677.7	757.7	742.2
IMF	65.3	67.3	57.0	47.7	47.5	37.4	25.4	19.0	12.9	8.3	5.2
World Bank/IDA	380.6	398.3	394.0	444.5	507.8	486.1	453.4	422.0	402.7	439.5	422.3
African Development Fund/Bank	182.8	169.8	159.9	168.5	179.8	205.8	173.7	182.4	185.5	208.7	210.2
EIB	3.9	7.9	7.3	13.0	13.0	12.4	9.9	9.8	8.7	9.4	10.0
Others	45.5	44.4	34.2	41.6	62.0	63.8	56.2	73.4	67.9	91.8	94.4
Bilateral	116.2	98.6	84.1	81.9	88.8	90.7	75.4	75.7	104.1	244.9	270.6
Paris Club official debt	30.2	25.8	24.0	25.2	24.3	23.2	23.6	19.2	16.6	32.0	16.1
Non-Paris Club official debt	86.1	72.7	60.1	56.7	64.4	67.5	51.8	56.5	87.5	212.9	254.5
of which: China, People's Republic	28.6	25.4	22.0	13.6	15.4	13.9	-	3.5	20.4	23.2	43.8
Libya	-	-	-	-	-	-	-	-	-	96.2	144.2
India	-	-	-	-	-	11.3	14.8	22.7	21.4	22.9	22.3
State-Guaranteed	-	-	-	-	-	-	-	-	-	64.3	304.4

Sources: World Bank, Chadian authorities, selected creditors and staff estimates.

The end-2011 external public debt stocks are estimated by staff based on World Bank Debt Reporting System (DRS) end-2010 debt stock, Chad's debt management office's estimates of 2011 disbursements and amortization, and Ministry of Economy and Planning estimates of 2011 project loan disbursements. The official external debt stock data underestimate the actual level of external debt. Most notably, the debt registry captures the loan from Libya but not the associated debt service, and it does not capture the loans from China for the state's stakes in the joint-venture refinery and in the Baoré cement factory, and other government-guaranteed debt. Also, project loan disbursements are recognized only after a long lag. Both text tables have discrepancies with corresponding fiscal or balance of payments flow estimates, giving rise to residuals in the sustainability analyses.

Status of Implementation of Debt Relief Initiatives

4. **Poor macroeconomic policy performance prevented Chad from reaching the completion point under the Enhanced Heavily Indebted Poor Countries (HIPC) Initiative**. Chad reached the Decision Point under the Enhanced HIPC Initiative in May 2001. Its inability to meet agreed fiscal targets and to implement a program under the IMF's Poverty Reduction and Growth Facility (PRGF) was a principal obstacle to debt relief. The 2005 PRGF arrangement expired in 2008 without any reviews being concluded. Subsequent efforts to resume the path toward debt relief with the support of IMF Staff-Monitored Programs (SMPs) were also hindered by fiscal slippages and, more recently, by the decision to borrow externally in large amounts and on non-concessional terms. Furthermore, Chad has not shown a satisfactory implementation record under the poverty reduction strategy and progress towards other completion point triggers[2] has either been slow or inconsistent (with early gains followed by backsliding).

[2] For a description of completion point triggers, see Chad, Decision Point Document for the Enhanced Heavily Indebted Poor Countries (HIPC) Initiative, May 4, 2001, pp. 29–31.

5. **Meeting the conditions for debt relief under the Enhanced HIPC Initiative and the Multilateral Debt Relief Initiative (MDRI) would help reduce external debt in half (in nominal terms)**. MDRI relief would cover the full stock of debt owed to three multilateral creditors (IDA, IMF, and the African Development Fund (AfDF)). In nominal terms, this could total over $1 billion and would imply a reduction in debt service of about $40 million per year, for about 30 years.[3]

Recent Developments in Public Domestic Debt

6. **The stock of public domestic debt has grown, mainly as a result of drawing on central bank statutory advances and the sale of over CFAF 100 billion of five-year savings bonds**. Chad's domestic debt is estimated at about CFAF 474.1 billion (9½ percent of GDP) at end-2011. The stock of public domestic debt includes central bank statutory advances (*avances statutaires*); rescheduled debt (*dettes conventionnées*); legal payment obligations (*engagements juridiques*); and two public bond issues. In addition, CFAF 56.4 billion represent estimated short-term treasury arrears.[4] In July 2011, the authorities completed a sale of over CFAF 100 billion of five year savings bonds with a 6 percent coupon.[5]

Text Table 2. Chad: Public Domestic Debt Stock at Year-End, 2006–2011
(Billions of CFA francs)

	2006	2007	2008	2009	2010	2011
Total	122.3	123.1	142.7	238.5	286.9	474.1
(Percent of GDP)	3.7	3.7	3.8	7.1	6.8	9.5
Central Bank Statutory Advances	-	17.0	21.6	141.6	208.6	216.9
Rescheduled debts	71.8	48.6	56.8	58.2	67.9	52.9
Treasury arrears [1]	24.8	26.1	41.1	25.7	3.1	56.4
Legal commitments	13.2	12.5	10.8	10.1	4.6	21.2
Standing payment orders	11.5	18.1	11.5	2.1	1.8	3.1
National Savings Bond	0.9	0.9	0.9	0.9	0.9	2.6
2011 Savings Bond	-	-	-	-	-	121.0

Source: Chadian authorities.
[1] The 2011 stock is estimated based on data at end-March 2012.

[3] Since the last remaining portion of Chad's debt owed to the IMF eligible for MDRI debt relief is due in July 2013, the IMF would provide no debt relief under the MDRI if the completion point was delayed beyond that time.

[4] The authorities made a significant effort to reduce domestic payment arrears in 2010, but payment arrears appear to have resurged in 2011.

[5] Sixty three percent of the issue was purchased by Chadian residents. Banks, local and regional, purchased two-thirds of the issue. With accrued interest, the stock of this public bond is estimated at CFAF 121 billion at end-December 2011.

Debt Burden Thresholds under the Debt Sustainability Framework

7. **For the purpose of defining debt burden thresholds under the Debt Sustainability Framework (DSF), Chad is a weak policy performer**. Chad's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) is 2.48 on average for 2007–2011 (on a scale from 1 to 6), down from 2.88 in 2005 to 2.43 in 2011.

Text Table 3. External Public Debt Burden Thresholds for "Weak Policy Performers" Under the Debt Sustainability Framework	
Present value of external debt in percent of	
GDP	30
Exports	100
Revenue	200
External debt service in percent of	
Exports	15
Revenue	18
Source: 2012 IDA Country Performance ratings (methodology and results).	

DSA ASSUMPTIONS

8. **Oil production and revenue:** Chad's medium- and long-term macroeconomic outlook is characterized by a temporary sharp increase in oil production in 2014–17 and a steady decline of oil production over the subsequent fifteen-year projection period (Box 1).[6]

- Oil production is expected to rise from 120,000 bpd in 2011 to about 180,000 bpd in 2015–17. This second oil boom will likely be temporary; proven reserves in the new fields are much smaller than that in the original Doba basin and will also likely be nearly exhausted around 2030. Hence, oil production and exports are projected to decline steadily to negligible levels beyond 2030.

[6] Oil production at the Doba oilfield (exploited by the Exxon-led consortium) started in 2003, reached its peak of 63 million barrels in 2005 and, absent other oil developments, will decline with annual output projected to become negligible beyond 2030. In 2011, oil production began at a second oil field, Bongor, operated by the CNPC (only one-third the size of the Doba field). At present, Bongor produces crude oil at a rate of about 15,000 barrels per day, as feedstock for the oil refinery, resulting in a near complete substitution of petroleum product imports. Exports of crude oil from Bongor are expected to start within a year and reach 60,000 bpd in 2014. Oil from another smaller operator (Griffiths, Canada) is expected to come on stream shortly thereafter.

- Chad's oil trades below the WEO reference price, reflecting a quality discount and transport cost.[7] For the medium term (five-year horizon) the price of Chadian oil is assumed to drop from $103.7 per barrel (all discounts included) in 2012 to about $83.6 per barrel in 2016–17, in line with the trend projected in the IMF's World Economic Outlook (WEO). From 2018 onward, the price is assumed to increase 3 percent per year in U.S. dollar terms (consistent with the assumption used by the IMF in long-term projections for other sub-Saharan African countries).

9. **Fiscal policy:** The analysis assumes a buildup of financial savings in the medium term and a sustained fiscal adjustment throughout the entire projection in transition to the post-oil era. The increase in oil production is expected to lead to about $4 billion more in oil revenues for the government by 2017 than projected in the 2011 Article IV report, helping to maintain oil revenues above 25 percent of non-oil GDP until 2016. This would be sufficient to generate savings of about $2 billion by 2019, if the non-oil primary deficit (NOPD) is reduced from about 29 percent of non-oil GDP in 2012 to below 10 percent of non-oil GDP by 2019. Over the longer term, it is assumed that dwindling oil revenues will be offset by a steady decline of total government primary spending from 27 percent of GDP in 2012 to 18 percent in 2032 and the primary balance will be adjusted gradually to reach a small surplus by 2032. This sustained process of adjustment is expected to be achieved mainly by: (i) increasing non-oil revenues (from about 10 percent of non-oil GDP at present to over 15 percent of non-oil GDP by 2032); (ii) reducing total investment outlays from over 20 percent of non-oil GDP to below 10 percent of non-oil GDP; and (iii) cutting recurrent spending, notably, by eliminating exceptional security transfers and subsidies to public enterprises (at present, jointly accounting for about 9 percent of non-oil GDP).

10. **The DSA analysis is conducted in two stages reflecting different external financing assumptions**. First, a **baseline (current policies) scenario** reflects the stated intentions of the authorities to implement the MFA on current terms, and therefore assumes that the full amount ($2 billion) is disbursed within a five-year (2012–17) horizon. The baseline scenario also features continuation of the borrowing policy on non-concessional terms beyond the medium term, with gross disbursement of about 3½ percent of GDP per year (approximately the same level as projected for the MFA implementation period). Second, the analysis aims at determining the conditions under which the implementation of MFA would be consistent with debt sustainability. This **alternative (prudent) scenario** is consistent with the medium-term fiscal framework recommended by staff and presented in the staff report.

[7] The discount was fairly constant (about $6.50) until recently when it dropped to about $2.50 due to the disruption of oil exports from South Sudan, which increased the demand for (a very similar) Chadian oil. This discount is expected to rebound over the near term to its normal level.

Box 1. Macroeconomic Assumptions, 2012–2032

Baseline Scenario:

Real GDP growth is driven by a sharp growth of oil production in the next five years and a steady decline in oil production over the following fifteen years. Large foreign-financed investments (see below) are assumed to raise **non-oil GDP growth** in the medium term (6.5 percent in 2013–17). Beyond that, non-oil GDP growth is assumed to stabilize at about 5.0 percent in the long term. **Inflation** is assumed to stabilize at 3 percent, consistent with the CEMAC convergence criterion. The **external current account** will turn into a significant surplus in 2014–17, with the increase in oil exports and the drop in imports related to the development of the new oil fields. With the decline in oil exports, the external current account is projected to turn into a deficit.

	2011	2012	2013	2014	2015	2016	2017	2011–17 Avg.	2018–32 Avg.
Real GDP growth (percent per year)	0.5	5.4	7.9	10.7	8.4	3.6	3.7	5.7	3.4
Oil	-2.5	-1.3	11.3	27.3	14.8	-4.6	-4.1	5.8	-9.6
Non-oil	1.4	7.1	7.1	6.6	6.5	6.2	6.0	5.8	4.8
Consumer price inflation (percent per year)	1.9	5.5	3.0	3.0	3.0	3.0	3.0	3.2	3.0
External current account balance (percent of GDP)	-1.0	-1.2	-3.9	0.1	4.8	2.4	1.7	0.4	-4.3
				(In percent of non-oil GDP)					
Government revenue and grants	49.3	43.1	38.9	40.1	43.9	39.0	35.5	41.4	23.4
Of which : oil revenue	35.3	29.6	25.0	25.4	28.7	23.8	20.1	26.8	6.4
Of which : grants	3.2	3.2	3.0	2.8	2.6	2.4	2.3	2.8	1.5
Government expenditure (commitment basis)	44.6	49.1	47.8	44.9	41.9	38.6	35.2	43.1	27.8
Overall fiscal balance (incl. grants; cash basis)	-2.4	-6.9	-11.9	-7.6	-0.7	-2.1	-1.9	-4.8	-4.2
Non-oil primary fiscal bal. (excl. grants; commitment basis)	-27.9	-29.0	-25.0	-21.6	-18.5	-15.5	-14.5	-21.7	-4.8
Memorandum items:									
Government deposits (in percent of non-oil GDP)	11.3	3.3	8.9	13.5	17.4	20.6	21.6	13.8	9.7
Chadian crude oil price (US$/barrel)	97.7	103.7	100.1	92.6	88.4	84.8	82.4	92.8	98.1

Sources: Chadian authorities; and IMF staff estimates and projections.

The **fiscal outlook** features an increase in oil revenues in the medium term and their decline in the long term. The authorities are assumed to reduce the NOPD to about 15 percent of non-oil GDP by 2017, and to generate savings up to $2 billion, needed as collateral for the MFA.

External financing: (i) a $2 billion MFA with the Eximbank of China on non-concessional terms and its implementation over 2012–17 (investments under the MFA are assumed to raise the non-oil GDP growth up by about one percentage point during the MFA implementation phase and by about 0.5 percentage point beyond 2017); (ii) external project financing (grants and loans) of 5.5 percent of non-oil GDP on average per year beyond 2017. Grants (mainly for social programs) are assumed to stay constant in real terms and will decline to about 1 percent of non-oil GDP by 2030. Loans are assumed to be increasingly from non-traditional (non-Paris Club) creditors, resulting in a decline of the average grant element of borrowing. In the absence of an IMF arrangement, HIPC and MDRI debt relief for which Chad is eligible are not taken into account.

Domestic financing: reimbursements of BEAC statutory advances by 2022 (as per the newly revised agreement) and the 2011 savings bond (*Emprunt obligataire)* by 2016. Additional domestic debt is assumed to be issued to meet the residual financing needs during the period when a portion of oil revenue is being set aside as collateral for the MFA. As a result, gross domestic debt rises from 9.5 percent of GDP in 2012 to about 11.5 percent in 2017.

Alternative Scenario:

Growth and **inflation** profile is broadly the same, with some differences in the first ten years reflecting mainly the implementation of the MFA over ten years (instead of five). The **fiscal outlook** features the same degree of consolidation over the medium term in NOPD terms (a maximum feasible rate of 2–3 percent of GDP per year). **External financing** assumptions: (i) MFA with no collateral requirement and on concessional terms (grant element of 35 percent); (ii) borrowing beyond MFA is also assumed to be on concessional terms with borrowing scaled back to approximately a half of the amount projected under the baseline. Since savings are not assumed to be related to the collateral requirements under the MFA, financing needs are met with lower additional domestic borrowing, such that gross **domestic debt** is projected to decline to 6 percent of GDP by 2017.

EXTERNAL DEBT DSA

Baseline (Current Policies) Scenario

11. **The evolution of external debt is driven by the large volume of project loans, including the MFA**. Going forward, new borrowing is expected to be dominated by non-concessional project financing, mainly from commercial or non-Paris Club official sources. This is predicated on the assumption that, absent a substantial improvement in macroeconomic and public financial management performance, traditional donors (multilaterals and Paris Club) will likely continue to provide roughly the same very low amount of concessional loans (compared to other SSA countries), mainly for humanitarian and development projects. During the period of MFA implementation, the average concessionality of new borrowing (assuming a discount rate of 3 percent) would turn negative and stay so through the projection period (Figure 1a).

12. **The comparison of the current baseline scenario with the baseline featured in the 2011 DSA reveals large differences reflecting a significant change in Chad's borrowing policy** (Figures 5–7). This change was anticipated by staffs in the 2011 DSA. It was presented in a customized alternative scenario featuring large non-concessional borrowing for public investment projects and drawing attention to the need for proper assessment of their consequences for debt sustainability. The signing of the MFA confirmed the importance of the risks anticipated by staffs in 2011.

13. **Under the baseline scenario, all external indebtedness indicators exceed their critical thresholds, indicating a high risk of debt distress**. The present value of external public and publicly-guaranteed debt exceeds the 30 percent of GDP threshold starting in 2016 through the end of the projection horizon (Figure 1b). With the projected steady decline in oil exports beyond the 2013–16 horizon, the PV of debt and debt service is expected to rise relative to exports, with the former projected to exceed the critical threshold starting in 2018, and the latter in 2026. Both indicators would remain above thresholds through the end of the projection period (Figures 1c and 1e). Similarly, the PV of debt and debt service in percent of revenues exceed their respective thresholds in 2022 and 2025, respectively (Figures 1d and 1f).

Stress Tests

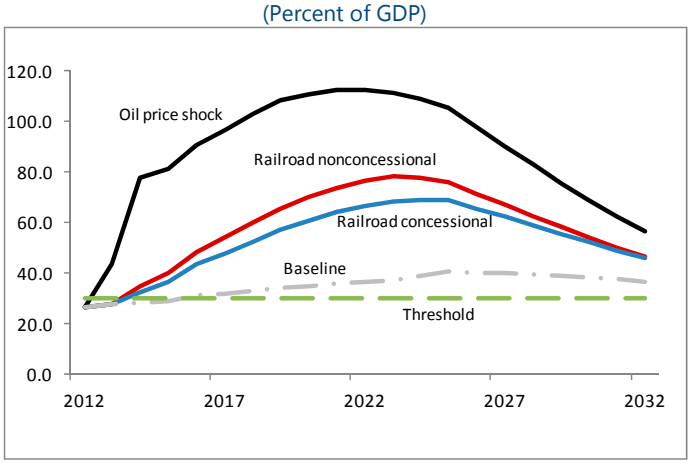
14. **Under the shock scenario assuming that key macroeconomic variables will remain at their ten-year average values, the debt burden indicators would quickly breach the sustainability thresholds**. This suggests that if the authorities were to incur current account deficits far higher than foreign direct investment (FDI) inflows in the oil sector (e.g., as they did in 2010, when they ran down their official foreign exchange

reserves), all three PV-based indicators would quickly breach the applicable thresholds by very high margins (Historical Scenario in Figure 1b, c, d; and Scenario A1 in Table 3b).

15. **Chad's external debt burden indicators are also highly sensitive to a negative shock to exports (e.g., an oil price shock)**. Across all indicative debt burden thresholds, the most extreme shock is a drop in export growth in 2013–14 (i.e., proportional to a two-standard-deviation lower oil price) (Most extreme shock in Figure 1 and B2 Bound Test in Table 3b); debt would be on a path breaching all indicative debt burden thresholds.

16. **Finally, a customized stress test shows that borrowing for other contemplated large capital projects (beyond the 2011 MFA) would increase even further the risk of debt distress**. In addition to the $2 billion MFA (signed in August 2011), the authorities have signed a letter of intent with a Chinese contractor for construction of an extensive railroad connection ($7 billion) linking through Chad the Cameroon and Sudan networks. A customized analysis by staff shows that if the railway project started in 2014 and was implemented over the next 10 years in addition to the projects covered in the 2011 MFA, and with no reduction in the baseline level of spending and borrowing, the PV of debt-to-GDP path would breach the 30 percent threshold in 2014 and peak in 2023 at 78 percent, implying a significantly increased risk of debt distress (Figure 1g). All critical thresholds would be breached under the usual commercial terms (5.5 percent interest rate, 5 years grace period, 20 years maturity) and also under the most concessional terms (36 percent grant element; 1.5 percent interest rate, 5 years grace period, 20 years maturity).

Figure 1g. Chad: Present Value of Public and Publicly-Guaranteed External Debt
(Percent of GDP)



Sources: Chadian authorities; and staff estimates and projections.

Alternative (Prudent) Scenario

This scenario is tailored to preserve debt sustainability. To this effect, it is assumed that $2 billion under the MFA would be disbursed over ten years, starting in late 2012. The MFA is assumed to be concessional, with a grant element of 35 percent. The analysis shows that the projected decline in exports beyond the medium term is the most stringent constraint on Chad's capacity to borrow. To make room for the disbursements under the MFA and, at the same time, ensure that the PV of debt-to-exports ratio remains below 100 percent, other borrowing would need to be scaled back to about one-third of the amount assumed in the baseline scenario even if the grant element of new borrowing is about 20 percent (Figure 3a). With this constraint satisfied, all other indicators remain below their critical thresholds. However, even in that case, shock scenarios point to a moderate risk of debt distress, which must be mitigated by prudent fiscal policy, and by maintaining an adequate savings cushion to ensure debt service capacity in case of a sudden loss of oil revenue.

PUBLIC DEBT DSA

17. **The consideration of domestic debt does not alter fundamentally the assessment of Chad's debt sustainability**. Given the moderate size of Chad's domestic debt, the baseline adjustment in the non-oil primary balance, and expected oil revenues, the public debt indicators are driven mainly by the external debt component, indicating a high risk of debt distress. The domestic debt component would increase from 9.5 percent of GDP in 2011 to 16.1 percent of GDP in 2032, reflecting the increased residual financing needs, as the portion of oil revenues used as collateral is not available for budget financing. Altogether, the public debt stock increases gradually from about 36 percent of GDP in 2011–12 to 53 percent of GDP in 2032.

18. **Stress tests confirm that the current fiscal stance is not sustainable**. Even with ample oil revenues, and assuming that financing could be secured, the resulting debt path would increase steeply, leading to an unmanageable debt and debt-service burden (Fixed Primary Balance Scenario in Figure 2). A temporary shock to real GDP growth in 2013-2014 would also impair public debt sustainability (Most Extreme Shock in Figure 2 and Bound Test B1 in Table 2a).

THE AUTHORITIES' VIEWS

19. **The authorities expressed interest in debt relief through the HIPC process**. They urged consideration of Chad's special post-conflict circumstances, the importance of public investment in infrastructure to long-term growth, and the need for debt relief to create fiscal space for additional development spending (including foreign-financed). At the same time, the authorities are determined to pursue an accelerated development strategy financed through foreign borrowing on non-concessional terms and secured through oil receipts.

20. **The authorities downplay concerns about the debt sustainability implications of the MFA**. They consider that new oil developments will generate a cash flow sufficient to meet debt obligations under the MFA. The authorities also believe that the contemplated investments will have significant positive economic externalities that would help sustain non-oil growth at a level higher than 5 percent per annum assumed in the DSA projections. The authorities intend to pursue their own debt sustainability analysis, with consultants, to assess the fiscal space for debt-financed public investment.

DEBT DISTRESS CLASSIFICATION AND CONCLUSIONS

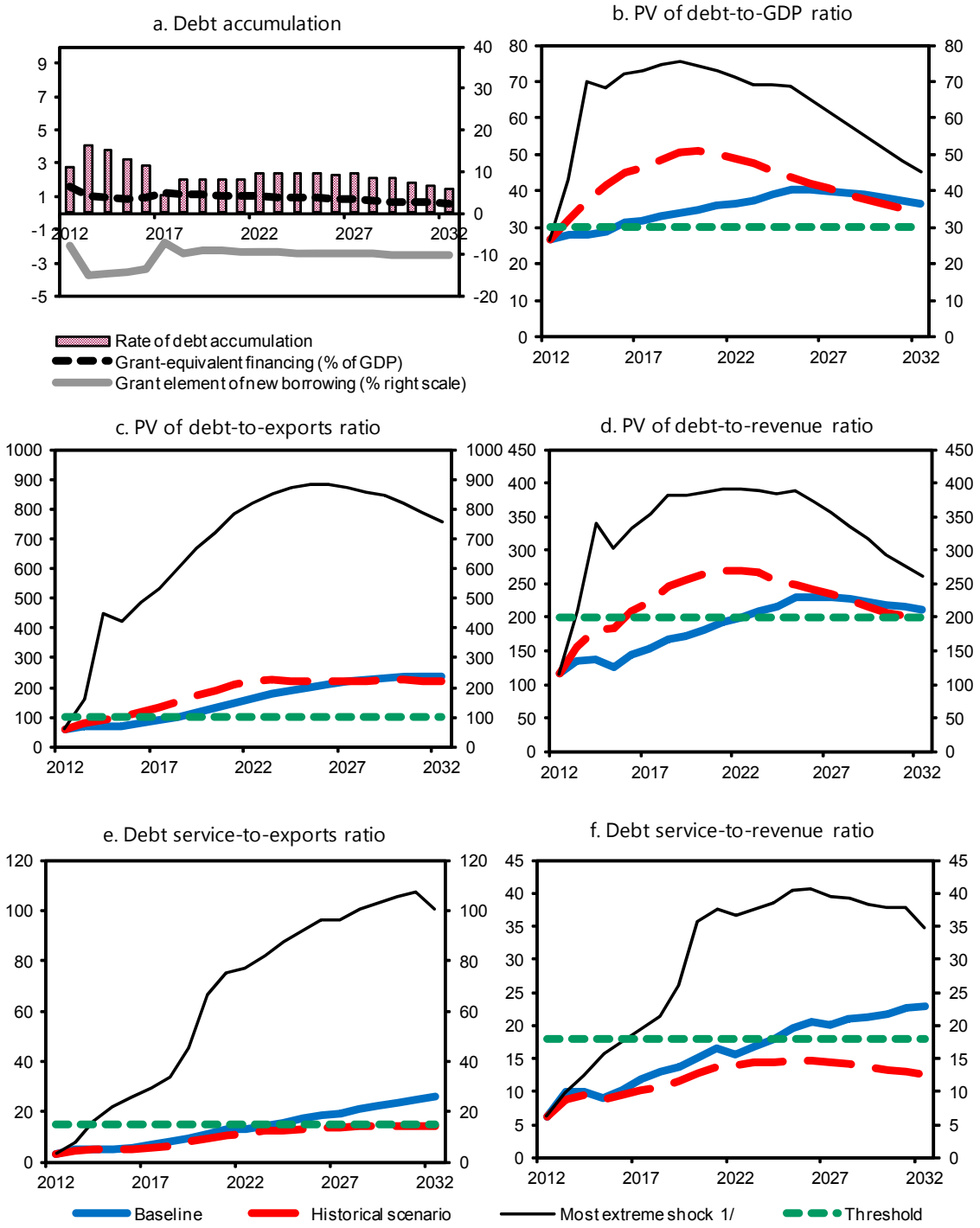
21. **In staff's assessment, despite improved oil production and revenue prospects in the next few years, Chad faces a high risk of debt distress**. The DSA suggests that under the baseline (current policies) scenario Chad's overall public and external debt dynamics are not sustainable. This is due to the significant increase in public external debt over the medium term, reflecting the implementation of the MFA on non-concessional terms and continuation with large amounts of non-concessional borrowing for other public investment projects. In this scenario, all public and publicly-guaranteed external debt and debt service indicators breach the critical thresholds. This conclusion holds even if the hypothetical growth impact of the associated projects is taken into account and the underlying fiscal position is assumed to be adjusted to help set aside funds needed to collateralize the disbursements under the MFA.

22. **An alternative (prudent) scenario shows that all indicative thresholds would be observed if the MFA were implemented over ten years and on concessional terms.** However, to make room for the MFA, all other borrowing would also need to be mostly concessional and its amount scaled back to about a half of the amount projected in the baseline scenario. Even in this case, the DSA underscores Chad's sensitivity to shocks, especially an oil price shock, and stress test results remain a source of concern.

23. **Progress toward the HIPC completion point (including the improvement in the CPIA score) would substantially reduce Chad's debt vulnerabilities**. However, given the current weak record for fiscal policy implementation and the non-concessional nature of the MFA with the Eximbank of China, prospects for re-embarking on the path to debt relief under the HIPC process are remote.

Baseline Scenario

Figure 1. Chad: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2012–2032 [1]

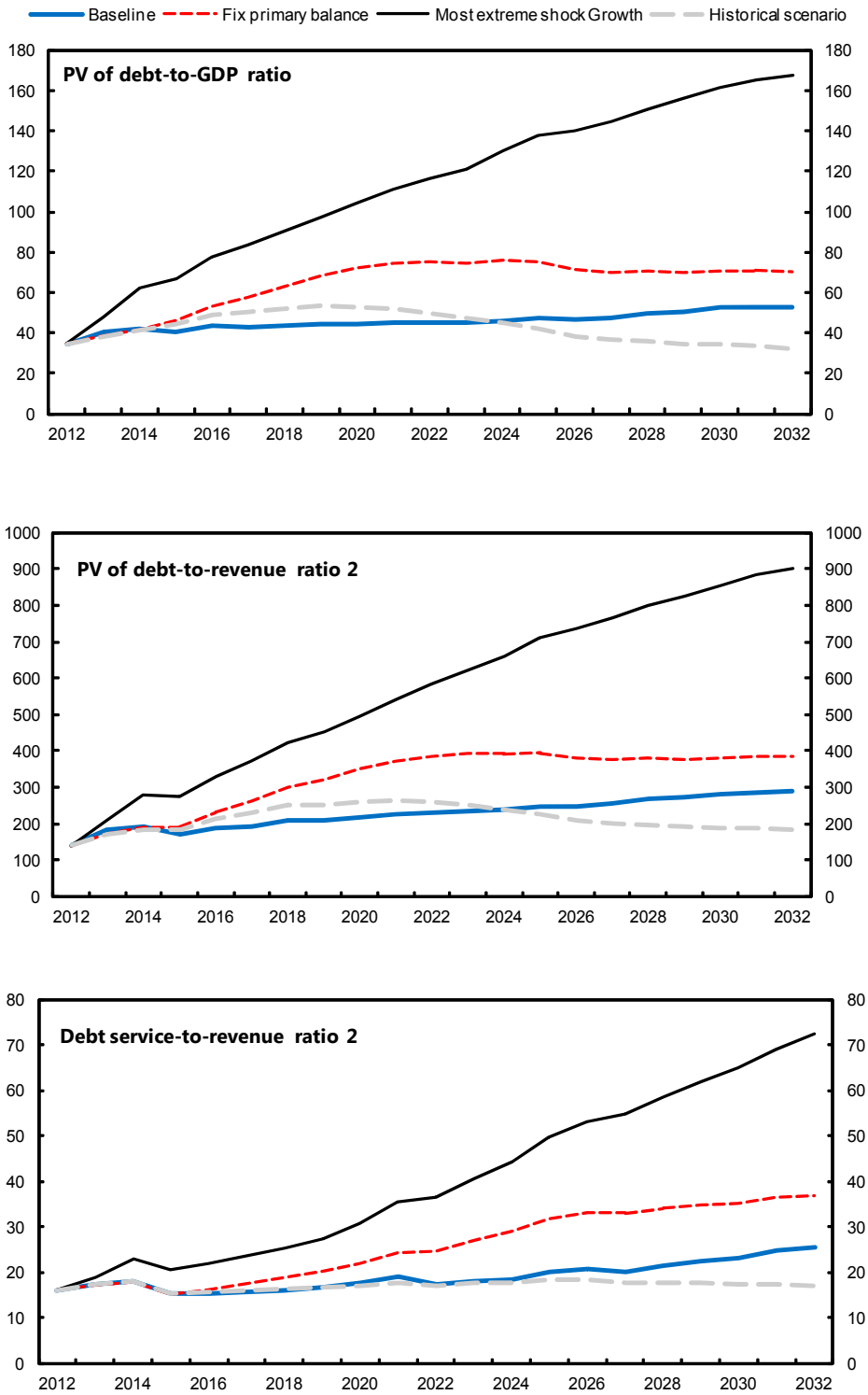


Sources: Country authorities; and staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to an Exports shock; in c. to an Exports shock; in d. to an Exports shock; in e. to an Exports shock and in figure f. to an Exports shock

Baseline Scenario

Figure 2. Chad: Indicators of Public Debt Under Alternative Scenarios, 2012–2032 [1]



Sources: Country authorities; and staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio in 2022.

[2] Revenue is defined inclusive of grants.

Baseline Scenario

Table 1. Chad Public Sector Debt Sustainability Framework, Baseline Scenario, 2009–2032

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average [5]	Standard Deviation [5]	2012	2013	2014	2015	2016	2017	2012–17 Average	2022	2032	2018–32 Average
Public sector debt [1]	30.5	32.8	35.9			35.6	41.0	42.2	40.4	42.6	42.5		44.8	52.8	
Of which: foreign-currency denominated	23.0	25.2	26.4			27.4	28.2	27.9	28.2	30.6	31.2		36.5	36.7	
Change in public sector debt	7.0	2.4	3.1			-0.4	5.5	1.2	-1.8	2.2	-0.1		0.0	-0.4	
Identified debt-creating flows	11.1	0.5	-7.7			0.8	2.0	-2.1	-4.7	-0.6	-1.1		1.3	0.9	
Primary deficit	9.4	4.7	-3.2	1.4	4.9	2.2	3.8	1.2	-2.7	-2.0	-2.0	0.1	1.4	2.2	2.
Revenue and grants	19.6	25.3	28.5			24.8	22.3	22.1	24.0	23.0	22.2		19.6	18.3	
Of which: grants	3.5	1.6	1.8			1.8	1.7	1.5	1.4	1.4	1.4		1.3	1.0	
Primary (noninterest) expenditure	29.0	29.9	25.3			27.0	26.2	23.3	21.4	21.0	20.2		21.0	20.5	
Automatic debt dynamics	1.7	-4.0	-4.5			-1.4	-1.9	-3.3	-2.0	1.4	0.9		-0.1	-1.3	
Contribution from interest rate/growth differential	-3.4	-6.8	-3.5			-5.1	-2.5	-3.5	-2.3	1.1	0.5		-0.1	-1.3	
Of which: contribution from average real interest rate	-3.2	-3.2	-3.3			-3.3	0.1	0.5	1.0	2.5	2.1		1.3	1.0	
Of which: contribution from real GDP growth	-0.2	-3.5	-0.2			-1.8	-2.6	-4.0	-3.3	-1.4	-1.5		-1.4	-2.3	
Contribution from real exchange rate depreciation	5.1	2.8	-1.0			3.7	0.6	0.2	0.2	0.2	0.3		...	...	
Other identified debt-creating flows	0.0	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-4.1	1.9	10.8			-1.1	3.5	3.3	2.9	2.8	1.0		-1.3	-1.3	
Other sustainability indicators															
PV of public sector debt	...	...	34.8			34.7	40.8	42.4	40.9	43.3	43.0		44.9	52.6	
Of which: foreign-currency denominated	...	...	25.3			26.5	28.0	28.1	28.7	31.3	31.7		36.7	36.4	
Of which: external	...	...	25.3			26.5	28.0	28.1	28.7	31.3	31.7		36.7	36.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need [2]	12.4	6.2	-2.2			6.2	7.7	5.2	1.0	1.5	1.5		4.8	6.8	
PV of public sector debt-to-revenue and grants ratio (percent)	...	...	122.3			140.0	182.5	191.8	170.0	188.2	194.3		229.4	287.5	
PV of public sector debt-to-revenue ratio (percent)	...	...	130.7			151.1	197.5	206.0	180.7	200.7	207.6		246.3	303.5	
Of which: external [3]	...	...	95.1			115.5	135.7	136.6	126.8	145.0	153.1		200.9	210.4	
Debt service-to-revenue and grants ratio (percent) [4]	8.8	4.1	3.6			16.2	17.2	18.1	15.4	15.4	15.6		17.4	25.4	
Debt service-to-revenue ratio (percent) [4]	10.7	4.4	3.9			17.5	18.7	19.4	16.4	16.4	16.7		18.7	26.8	
Primary deficit that stabilizes the debt-to-GDP ratio	2.4	2.3	-6.3			2.6	-1.6	0.0	-0.8	-4.2	-1.9		1.4	2.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (percent)	0.9	13.0	0.5	7.9	10.7	5.4	7.9	10.7	8.4	3.6	3.7	6.6	3.3	4.5	3.
Average nominal interest rate on forex debt (percent)	1.4	1.0	2.2	1.2	0.4	3.5	3.5	3.7	3.9	4.1	4.2	3.8	4.2	4.2	4.
Average real interest rate on domestic debt (percent)	19.0	-5.8	...	2.6	11.3	-2.4	3.2	2.2	3.3	7.1	5.7	3.2	3.3	2.2	3.
Real exchange rate depreciation (percent, + indicates depreciation)	30.0	15.9	-4.4	-2.9	14.7	16.9	...	...	...	...	...	...	...	...	-
Inflation rate (GDP deflator, percent)	-11.4	11.9	17.5	7.3	9.8	7.4	1.8	2.5	1.3	-2.3	-1.1	1.6	1.2	2.3	1.
Growth of real primary spending (deflated by GDP deflator, percent)	0.3	0.2	-0.2	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.
Grant element of new external borrowing (percent)	...	...	...	...	...	-7.8	-15.1	-14.6	-14.3	-13.6	-7.1	-12.1	-9.4	-10.0	-

Sources: Country authorities; and staff estimates and projections.

[1] [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenue excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Baseline Scenario

Table 2. Chad: Sensitivity Analysis for Key Indicators of Public Debt, 2012–2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of debt-to-GDP ratio								
Baseline	35	41	42	41	43	43	45	53
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	35	38	41	45	49	51	50	33
A2. Primary balance is unchanged from 2012	35	39	42	46	53	58	75	70
A3. Permanently lower GDP growth [1]	35	42	46	47	53	57	89	203
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013-2014	35	48	63	67	77	84	116	167
B2. Primary balance is at historical average minus one standard deviation in 2013-2014	35	44	51	49	51	51	54	60
B3. Combination of B1-B2 using one-half standard deviation shocks	35	43	51	52	59	62	79	110
B4. One-time 30 percent real depreciation in 2013	35	52	53	51	55	55	60	74
B5. 10 percent of GDP increase in other debt-creating flows in 2013	35	52	53	51	54	53	56	63
PV of debt-to-revenue Ratio [2]								
Baseline	140	183	192	170	188	194	229	287
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	140	170	185	185	213	230	259	183
A2. Primary balance is unchanged from 2012	140	174	189	191	232	262	384	383
A3. Permanently lower GDP growth [1]	140	189	207	194	229	254	446	1079
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013-2014	140	213	278	275	330	371	582	902
B2. Primary balance is at historical average minus one standard deviation in 2013-2014	140	195	230	203	224	232	274	331
B3. Combination of B1-B2 using one-half standard deviation shocks	140	190	227	216	253	277	402	595
B4. One-time 30 percent real depreciation in 2013	140	233	241	214	238	248	306	405
B5. 10 percent of GDP increase in other debt-creating flows in 2013	140	234	240	212	233	241	285	342
Debt service-to-revenue ratio [2]								
Baseline	16	17	18	15	15	16	17	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	17	18	15	16	16	17	17
A2. Primary balance is unchanged from 2012	16	17	18	15	16	18	25	37
A3. Permanently lower GDP growth [1]	16	18	19	17	17	18	28	78
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013-2014	16	19	23	21	22	23	37	72
B2. Primary balance is at historical average minus one standard deviation in 2013-2014	16	17	19	17	17	17	21	30
B3. Combination of B1-B2 using one-half standard deviation shocks	16	18	20	18	18	19	26	49
B4. One-time 30 percent real depreciation in 2013	16	19	22	19	20	21	27	45
B5. 10 percent of GDP increase in other debt-creating flows in 2013	16	17	20	17	17	17	22	31

Sources: Country authorities; and staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

[2] Revenues are defined inclusive of grants.

Baseline Scenario

Table 3a. Chad: External Debt Sustainability Framework, Baseline Scenario, 2009–2032 [1]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical [6] Average	Standard [6] Deviation	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012–2017 Average	2022	2032	2018–2032 Average
External debt (nominal) [1]	**23.0**	**25.2**	**26.4**			**27.4**	**28.2**	**27.9**	**28.2**	**30.6**	**31.2**		**36.5**	**36.7**	
Of which: public and publicly guaranteed (PPG)	23.0	25.2	26.4			27.4	28.2	27.9	28.2	30.6	31.2		36.5	36.7	
Change in external debt	2.1	2.2	1.2			1.0	0.9	-0.3	0.3	2.4	0.6		0.8	-1.0	
Identified net debt-creating flows	-1.5	4.6	-6.5			-1.8	-0.7	-1.2	-1.1	0.1	0.3		2.2	2.5	
Non-interest current account deficit	**10.0**	**31.1**	**0.5**	**21.4**	**30.1**	**0.3**	**3.1**	**-1.0**	**-5.8**	**-3.5**	**-2.9**		**3.4**	**3.4**	2.8
Deficit in balance of goods and services	11.9	30.7	3.4			2.6	3.2	-0.7	-6.2	-3.2	-2.5		6.2	7.1	
Exports	44.1	45.0	44.7			43.9	40.2	40.2	41.7	38.2	35.2		22.3	15.4	
Imports	56.0	75.6	48.0			46.5	43.4	39.5	35.5	35.0	32.7		28.5	22.5	
Net current transfers (negative = inflow)	-7.3	-7.5	-5.4	-5.4	1.2	-8.4	-4.8	-4.4	-4.2	-4.3	-4.4		-4.5	-3.8	-4.3
Of which: official	-1.0	-0.5	-0.3			-0.3	-0.4	-0.3	-0.3	-0.4	-0.4		-0.5	-0.7	
Other current account flows (negative = net inflow)	5.4	7.9	2.5			6.1	4.7	4.1	4.6	4.0	4.0		1.7	0.2	
Net FDI (negative = inflow)	**-15.6**	**-22.7**	**-2.7**	**-14.6**	**14.2**	**-1.6**	**-2.7**	**1.6**	**5.8**	**3.5**	**3.1**		**-1.6**	**-0.9**	-0.9
Endogenous debt dynamics [2]	**4.2**	**-3.7**	**-4.4**			**-0.5**	**-1.1**	**-1.8**	**-1.1**	**0.2**	**0.1**		**0.3**	**-0.1**	
Contribution from nominal interest rate	0.4	0.2	0.5			0.9	0.9	0.9	1.0	1.2	1.2		1.5	1.5	
Contribution from real GDP growth	-0.2	-2.5	-0.1			-1.4	-2.0	-2.7	-2.1	-1.0	-1.1		-1.1	-1.6	
Contribution from price and exchange rate changes	4.0	-1.4	-4.7			…	…	…	…	…	…		…	…	
Residual (3-4) [3]	**3.6**	**-2.4**	**7.7**			**2.8**	**1.6**	**0.9**	**1.4**	**2.3**	**0.3**		**-1.3**	**-3.5**	
Of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt [4]	…	…	25.3			26.5	28.0	28.1	28.7	31.3	31.7		36.7	36.4	
Percent of exports	…	…	56.7			60.5	69.7	70.1	68.7	81.9	90.1		164.1	236.3	
PV of PPG external debt	**…**	**…**	**25.3**			**26.5**	**28.0**	**28.1**	**28.7**	**31.3**	**31.7**		**36.7**	**36.4**	
Percent of exports	**…**	**…**	**56.7**			**60.5**	**69.7**	**70.1**	**68.7**	**81.9**	**90.1**		**164.1**	**236.3**	
Percent of government revenues	**…**	**…**	**95.1**			**115.6**	**135.7**	**136.6**	**126.8**	**145.0**	**153.1**		**200.9**	**210.4**	
Debt service-to-exports ratio (percent)	**2.1**	**1.6**	**2.3**			**3.3**	**5.1**	**5.1**	**5.0**	**5.8**	**6.9**		**12.8**	**25.9**	
PPG debt service-to-exports ratio (percent)	**2.1**	**1.6**	**2.3**			**3.3**	**5.1**	**5.1**	**5.0**	**5.8**	**6.9**		**12.8**	**25.9**	
PPG debt service-to-revenue ratio (percent)	**5.7**	**3.1**	**3.9**			**6.3**	**10.0**	**9.9**	**9.1**	**10.3**	**11.8**		**15.7**	**23.0**	
Total gross financing need (billions of U.S. dollars)	-0.3	0.8	-0.1			0.0	0.3	0.3	0.3	0.3	0.4		0.8	1.9	
Non-interest current account deficit that stabilizes debt ratio	7.9	28.8	-0.7			-0.6	2.2	-0.7	-6.0	-5.9	-3.6		2.6	4.4	
Key macroeconomic assumptions															
Real GDP growth (percent)	0.9	13.0	0.5	7.9	10.7	5.4	7.9	10.7	8.4	3.6	3.7	6.6	3.3	4.5	3.4
GDP deflator in US dollar terms (change in percent)	-16.1	6.6	23.2	12.3	12.1	-2.3	-0.4	2.2	0.8	-2.7	-1.7	-0.7	1.2	2.3	1.3
Effective interest rate (percent) [5]	1.4	1.0	2.2	1.2	0.4	3.5	3.5	3.7	3.9	4.1	4.2	3.8	4.2	4.2	4.2
Growth of exports of G&S (U.S. dollar terms, percent)	-30.9	23.0	22.9	49.8	84.1	1.2	-1.6	13.1	13.7	-7.8	-5.9	2.1	-2.7	4.5	-0.9
Growth of imports of G&S (U.S. dollar terms, percent)	-1.5	62.8	-21.4	27.6	53.5	-0.3	0.3	3.0	-1.6	-0.8	-4.7	-0.7	2.2	4.3	2.1
Grant element of new public sector borrowing (percent)	…	…	…	…	…	-7.8	-15.1	-14.6	-14.3	-13.6	-7.1	-12.1	-9.4	-10.0	-9.6
Government revenue (excluding grants, percent of GDP)	16.1	23.6	26.6			23.0	20.6	20.6	22.6	21.6	20.7		18.2	17.3	18.1
Aid flows (billions of U.S. dollars) [7]	0.3	0.3	0.3			0.2	0.2	0.2	0.2	0.2	0.2		0.3	0.3	
Of which: Grants	0.2	0.1	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.3	
Of which: Concessional loans	0.1	0.1	0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (percent of GDP) [8]	…	…	…			1.6	1.0	0.9	0.9	0.9	1.2		1.0	0.6	0.9
Grant-equivalent financing (percent of external financing) [8]	…	…	…			32.1	16.7	16.7	17.4	18.2	32.5		19.5	11.7	17.0
Memorandum items:															
Nominal GDP (billions of U.S. dollars)	7.1	8.6	10.6			10.9	11.7	13.3	14.5	14.6	14.9		17.4	29.6	
Nominal dollar GDP growth	-15.4	20.5	23.8			3.0	7.4	13.2	9.3	0.7	2.0	5.9	4.5	6.9	4.7
PV of PPG external debt (billions of U.S. dollars)			2.5			2.8	3.3	3.7	4.2	4.6	4.7		6.4	10.8	
(PVt-PVt-1)/GDPt-1 (percent)						2.8	4.1	3.8	3.2	2.8	1.1	3.0	2.4	1.5	2.1
Gross workers' remittances (billions of U.S. dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (percent of GDP + remittances)	…	…	25.3			26.5	28.0	28.1	28.7	31.3	31.7		36.7	36.4	
PV of PPG external debt (percent of exports + remittances)	…	…	56.7			60.5	69.7	70.1	68.7	81.9	90.1		164.1	236.3	
Debt service of PPG external debt (percent of exports + remittances)	…	…	2.3			3.3	5.1	5.1	5.0	5.8	6.9		12.8	25.9	

Sources: Country authorities; and staff estimates and projections.

[1] Includes both public and private sector external debt.

[2] Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate chan

[4] Assumes that PV of private sector debt is equivalent to its face value.

[5] Current-year interest payments divided by previous period debt stock.

[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7] Defined as grants, concessional loans, and debt relief.

[8] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Baseline Scenario

Table 3b. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032

(Percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	27	28	28	29	31	32	**37**	36
A. Alternative scenarios								
A1. Key variables at their historical averages in 2012–2032 [1]	27	32	37	42	45	46	**49**	34
A2. New public sector loans on less favorable terms in 2012–2032 2	27	29	30	31	34	36	**45**	52
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	27	31	35	36	39	40	**46**	46
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	27	43	70	68	72	73	**71**	45
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	27	28	28	29	32	32	**37**	37
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	27	31	29	30	32	33	**37**	36
B5. Combination of B1-B4 using one-half standard deviation shocks	27	18	7	8	10	10	**19**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	27	40	40	41	45	45	**52**	52
PV of debt-to-exports ratio								
Baseline	61	70	70	69	82	90	**164**	236
A. Alternative scenarios								
A1. Key variables at their historical averages in 2012–2032 [1]	61	80	93	100	118	131	**220**	221
A2. New public sector loans on less favorable terms in 2012–2032 2	61	72	74	74	90	101	**201**	338
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	61	70	70	69	82	90	**164**	236
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	61	161	449	422	486	535	**821**	758
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	61	70	70	69	82	90	**164**	236
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	61	78	73	71	84	93	**167**	237
B5. Combination of B1-B4 using one-half standard deviation shocks	61	41	15	19	25	28	**81**	197
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	61	70	70	69	82	90	**164**	236
PV of debt-to-revenue ratio								
Baseline	116	136	137	127	145	153	**201**	210
A. Alternative scenarios								
A1. Key variables at their historical averages in 2012–2032 [1]	116	157	181	184	209	223	**269**	197
A2. New public sector loans on less favorable terms in 2012–2032 [2]	116	139	144	137	159	171	**246**	301
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	116	150	172	160	183	193	**253**	265
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	116	209	340	302	334	353	**391**	262
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	116	135	138	128	147	155	**203**	213
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	116	151	141	131	149	158	**204**	211
B5. Combination of B1-B4 using one-half standard deviation shocks	116	85	32	37	48	50	**105**	187
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	116	194	195	181	207	219	**287**	300

Baseline Scenario

Table 3b. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032 (continued)

(Percent)

Debt service-to-exports ratio								
Baseline	3	5	5	5	6	7	**13**	26
A. Alternative scenarios								
A1. Key variables at their historical averages in 2012–2032 [1]	3	5	5	5	5	6	**11**	14
A2. New public sector loans on less favorable terms in 2012–2032 [2]	3	5	5	5	6	7	**15**	35
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	3	5	5	5	6	7	**13**	26
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	3	8	16	22	25	29	**77**	101
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	3	5	5	5	6	7	**13**	26
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	3	5	5	5	6	7	**13**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	3	5	4	3	4	4	**4**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	3	5	5	5	6	7	**13**	26
Debt service-to-revenue ratio								
Baseline	6	10	10	9	10	12	**16**	23
A. Alternative scenarios								
A1. Key variables at their historical averages in 2012–2032 [1]	6	9	9	9	9	10	**14**	13
A2. New public sector loans on less favorable terms in 2012–2032 [2]	6	10	9	9	10	12	**19**	31
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	6	11	13	12	13	15	**20**	29
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	6	10	12	16	18	19	**37**	35
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	6	10	10	9	10	12	**16**	23
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	6	10	10	9	10	12	**16**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	6	10	8	6	7	8	**5**	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	6	14	14	13	15	17	**22**	33
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	-14	-14	-14	-14	-14	-14	**-14**	-14

Sources: Country authorities; and staff estimates and projections.

[1] Variables include real GDP growth, growth of GDP deflator (U.S. dollar terms), non-interest current account in percent of GDP, and non-debt-creating flows.

[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

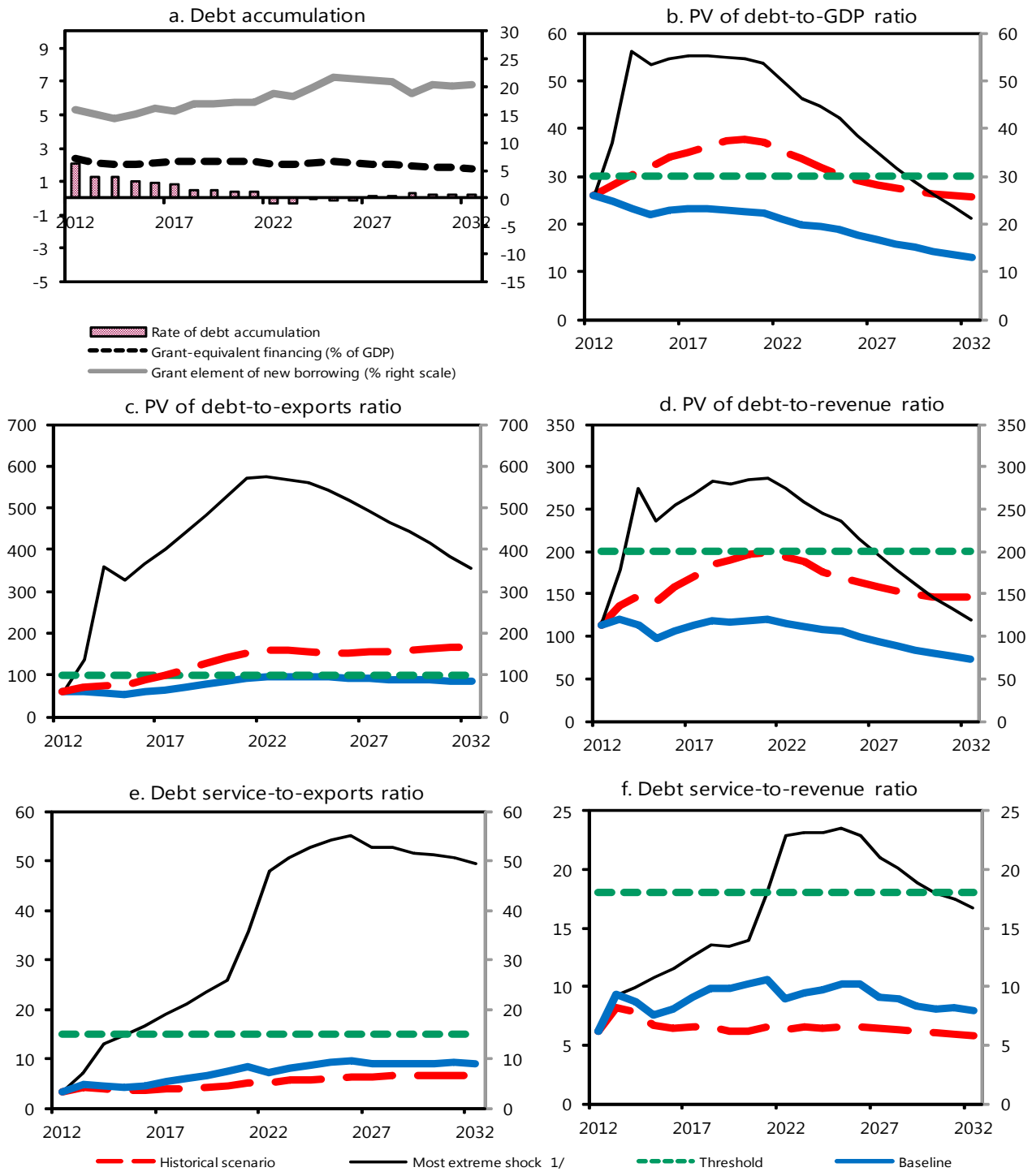
[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Alternative Scenario

Figure 3. Chad: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2012–2032 [1]

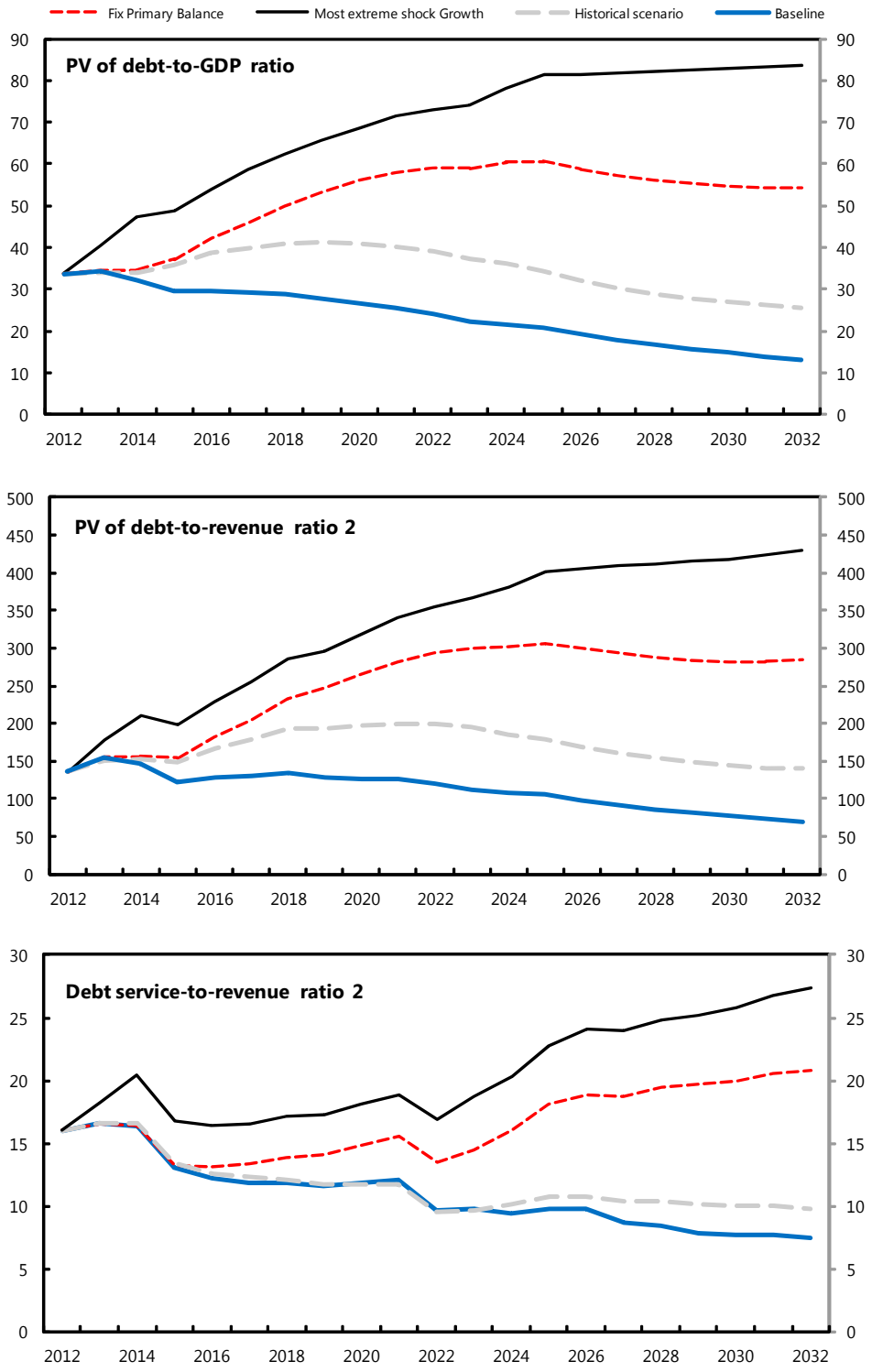


Sources: Country authorities; and staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to an Exports shock; in c. to an Exports shock; in d. to an Exports shock; in e. to an Exports shock and in figure f. to an Exports shock.

Alternative Scenario

Figure 4. Chad: Indicators of Public Debt Under Alternative Scenarios, 2012–2032 [1]



Sources: Country authorities; and staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio in 2022.

[2] Revenue is defined inclusive of grants.

Alternative Scenario

Table 4a. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009–2032

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average [5]	Standard Deviation [5]	2012	2013	2014	2015	2016	2017	2012–17 Average	2022	2032	2018–32 Average
Public sector debt [1]	30.5	32.8	35.9			35.4	36.4	34.3	31.7	32.1	32.2		27.5	16.7	
Of which: foreign-currency denominated	23.0	25.2	26.4			27.6	26.8	25.2	24.3	25.5	26.1		24.7	16.3	
Change in public sector debt	7.0	2.4	3.1			-0.5	1.0	-2.1	-2.6	0.4	0.1		-1.9	-0.7	
Identified debt-creating flows	11.1	0.5	-7.7			0.9	0.0	-3.4	-5.9	-2.5	-1.7		-2.1	-1.3	
Primary deficit	9.4	4.7	-3.2	1.4	4.9	2.2	2.1	-0.3	-3.9	-3.3	-2.0	-0.9	-1.5	-0.5	-0.5
Revenue and grants	19.6	25.3	28.5			24.9	22.4	22.2	24.1	23.2	22.5		20.1	19.1	
Of which: grants	3.5	1.6	1.8			1.8	1.7	1.7	1.7	1.8	1.9		1.8	1.4	
Primary (noninterest) expenditure	29.0	29.9	25.3			27.1	24.4	21.9	20.2	20.0	20.5		18.6	18.6	
Automatic debt dynamics	1.7	-4.0	-4.5			-1.3	-2.0	-3.2	-1.9	0.8	0.3		-0.6	-0.8	
Contribution from interest rate/growth differential	-3.4	-6.8	-3.5			-5.0	-2.6	-3.3	-2.1	0.6	0.1		-0.6	-0.8	
Of which: contribution from average real interest rate	-3.2	-3.2	-3.3			-3.2	-0.1	0.1	0.5	1.6	1.2		0.3	-0.1	
Of which: contribution from real GDP growth	-0.2	-3.5	-0.2			-1.8	-2.5	-3.5	-2.6	-1.0	-1.1		-0.9	-0.7	
Contribution from real exchange rate depreciation	5.1	2.8	-1.0			3.7	0.6	0.2	0.2	0.2	0.2		...	...	
Other identified debt-creating-flows	0.0	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-4.1	1.9	10.8			-1.4	1.0	1.3	3.3	2.9	1.8		0.2	0.6	
Other sustainability indicators															
PV of public sector debt	...	...	34.8			33.8	34.5	32.3	29.5	29.5	29.3		23.9	13.3	
Of which: foreign-currency denominated	...	...	25.3			26.0	24.9	23.1	22.1	22.8	23.2		21.1	12.9	
Of which: external	...	...	25.3			26.0	24.9	23.1	22.1	22.8	23.2		21.1	12.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need [2]	12.4	6.2	-2.2			6.2	5.8	3.4	-0.8	-0.4	0.7		0.4	0.9	
PV of public sector debt-to-revenue and grants ratio (percent)	...	...	122.3			135.5	154.3	145.5	122.3	127.2	130.4		119.0	69.6	
PV of public sector debt-to-revenue ratio (percent)	...	...	130.7			146.3	166.8	157.3	131.3	137.7	142.2		131.1	75.3	
Of which: external [3]	...	...	95.1			112.5	120.2	112.7	98.3	106.6	112.7		115.4	73.1	
Debt service-to-revenue and grants ratio (percent) [4]	8.8	4.1	3.6			16.0	16.6	16.4	13.0	12.2	11.8		9.6	7.4	
Debt service-to-revenue ratio (percent) [4]	10.7	4.4	3.9			17.3	17.9	17.7	14.0	13.2	12.9		10.6	8.0	
Primary deficit that stabilizes the debt-to-GDP ratio	2.4	2.3	-6.3			2.7	1.0	1.8	-1.4	-3.6	-2.1		0.4	0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (percent)	0.9	13.0	0.5	7.9	10.7	5.4	7.7	10.5	8.2	3.4	3.5	6.4	3.2	4.5	3.4
Average nominal interest rate on forex debt (percent)	1.4	1.0	2.2	1.2	0.4	3.3	2.9	2.8	2.7	2.7	2.7	2.8	2.1	1.8	2.0
Average real interest rate on domestic debt (percent)	19.0	-5.8	...	2.6	11.3	-1.9	3.1	2.1	3.3	7.1	5.7	3.2	3.3	2.1	3.2
Real exchange rate depreciation (percent, + indicates depreciation)	30.0	15.9	-4.4	-2.9	14.7	16.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, percent)	-11.4	11.9	17.5	7.3	9.8	6.9	1.9	2.7	1.5	-2.2	-1.1	1.6	1.2	2.4	1.3
Growth of real primary spending (deflated by GDP deflator, percent)	0.3	0.2	-0.2	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (percent)	...	...	...	...	...	16.0	15.2	14.4	15.1	16.2	15.5	15.4	18.9	20.3	...

Sources: Country authorities; and staff estimates and projections.

[1] [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenue excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Alternative Scenario

Table 5a. Chad: Sensitivity Analysis for Key Indicators of Public Debt, 2012–2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of debt-to-GDP ratio								
Baseline	34	35	32	30	30	29	24	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	34	34	34	36	39	40	39	26
A2. Primary balance is unchanged from 2012	34	35	35	37	42	46	59	54
A3. Permanently lower GDP growth [1]	34	36	35	34	37	39	54	108
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013–2014	34	40	47	49	54	59	73	84
B2. Primary balance is at historical average minus one standard deviation in 2013–2014	34	38	41	38	38	38	33	20
B3. Combination of B1-B2 using one-half standard deviation shocks	34	38	42	41	44	46	50	50
B4. One-time 30 percent real depreciation in 2013	34	45	42	38	38	38	32	20
B5. 10 percent of GDP increase in other debt-creating-flows in 2013	34	43	40	37	37	37	32	19
PV of debt-to-revenue ratio [2]								
Baseline	136	154	145	122	127	130	119	70
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	136	151	153	149	167	178	198	140
A2. Primary balance is unchanged from 2012	136	155	156	154	182	204	293	285
A3. Permanently lower GDP growth [1]	136	159	157	140	157	174	265	541
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013–2014	136	178	210	199	228	255	355	430
B2. Primary balance is at historical average minus one standard deviation in 2013–2014	136	171	187	158	165	170	162	102
B3. Combination of B1-B2 using one-half standard deviation shocks	136	168	186	168	186	201	247	260
B4. One-time 30 percent real depreciation in 2013	136	201	187	158	164	169	160	103
B5. 10 percent of GDP increase in other debt-creating-flows in 2013	136	194	182	154	161	165	157	98
Debt service-to-revenue ratio [2]								
Baseline	16	17	16	13	12	12	10	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	17	17	13	13	12	9	10
A2. Primary balance is unchanged from 2012	16	17	16	13	13	13	13	21
A3. Permanently lower GDP growth [1]	16	17	17	14	14	14	14	29
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2013–2014	16	18	20	17	16	16	17	27
B2. Primary balance is at historical average minus one standard deviation in 2013–2014	16	17	17	14	13	13	11	10
B3. Combination of B1-B2 using one-half standard deviation shocks	16	17	18	15	15	14	14	18
B4. One-time 30 percent real depreciation in 2013	16	18	20	16	16	16	14	13
B5. 10 percent of GDP increase in other debt-creating-flows in 2013	16	17	17	14	13	13	13	10

Sources: Country authorities; and staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

[2] Revenue is defined inclusive of grants.

Alternative Scenario

Table 6a. Chad: External Debt Sustainability Framework, Baseline Scenario, 2009–2032 [1]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical [6] Average	Standard [6] Deviation	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012–2017 Average	2022	2032	2018–2032 Average
External debt (nominal) [1]	**23.0**	**25.2**	**26.4**			**27.6**	**26.8**	**25.2**	**24.3**	**25.5**	**26.1**		**24.7**	**16.3**	
Of which: public and publicly guaranteed (PPG)	23.0	25.2	26.4			27.6	26.8	25.2	24.3	25.5	26.1		24.7	16.3	
Change in external debt	2.1	2.2	1.2			1.2	-0.8	-1.6	-0.8	1.1	0.7		-1.3	-0.6	
Identified net debt-creating-flows	-1.5	4.6	-6.5			-1.9	-1.3	-1.6	-0.8	-0.7	0.2		1.1	2.0	
Non-interest current account deficit	**10.0**	**31.1**	**0.5**	**21.4**	**30.1**	**0.3**	**2.6**	**-1.4**	**-6.0**	**-3.8**	**-2.9**		**3.0**	**3.3**	2.6
Deficit in balance of goods and services	11.9	30.7	3.4			2.6	2.7	-1.1	-6.6	-3.6	-2.5		5.8	6.8	
Exports	44.1	45.0	44.7			44.1	40.4	40.4	42.0	38.5	35.5		22.4	15.4	
Imports	56.0	75.6	48.0			46.7	43.1	39.3	35.4	34.9	33.0		28.2	22.2	
Net current transfers (negative = inflow)	-7.3	-7.5	-5.4	-5.4	1.2	-8.4	-4.8	-4.4	-4.2	-4.3	-4.4		-4.5	-3.8	-4.3
Of which: official	-1.0	-0.5	-0.3			-0.4	-0.4	-0.3	-0.3	-0.4	-0.4		-0.5	-0.7	
Other current account flows (negative = net inflow)	5.4	7.9	2.5			6.1	4.7	4.1	4.8	4.1	4.1		1.7	0.2	
Net FDI (negative = inflow)	**-15.6**	**-22.7**	**-2.7**	**-14.6**	**14.2**	**-1.6**	**-2.7**	**1.6**	**6.5**	**3.2**	**3.2**		**-1.6**	**-0.9**	-0.9
Endogenous debt dynamics [2]	**4.2**	**-3.7**	**-4.4**			**-0.5**	**-1.2**	**-1.8**	**-1.3**	**-0.2**	**-0.2**		**-0.3**	**-0.4**	
Contribution from nominal interest rate	0.4	0.2	0.5			0.9	0.7	0.7	0.6	0.7	0.7		0.5	0.3	
Contribution from real GDP growth	-0.2	-2.5	-0.1			-1.4	-2.0	-2.5	-1.9	-0.8	-0.9		-0.8	-0.7	
Contribution from price and exchange rate changes	4.0	-1.4	-4.7			...	...	...	...	...	...		...	...	
Residual (3-4) [3]	**3.6**	**-2.4**	**7.7**			**3.0**	**0.5**	**0.0**	**-0.1**	**1.9**	**0.5**		**-2.4**	**-2.6**	
Of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt [4]	...	...	25.3			26.0	24.9	23.1	22.1	22.8	23.2		21.1	12.9	
Percent of exports	...	...	56.7			58.9	61.5	57.2	52.6	59.3	65.4		94.0	83.8	
PV of PPG external debt	**...**	**...**	**25.3**			**26.0**	**24.9**	**23.1**	**22.1**	**22.8**	**23.2**		**21.1**	**12.9**	
Percent of exports	**...**	**...**	**56.7**			**58.9**	**61.5**	**57.2**	**52.6**	**59.3**	**65.4**		**94.0**	**83.8**	
Percent of government revenues	**...**	**...**	**95.1**			**112.5**	**120.2**	**112.7**	**98.3**	**106.6**	**112.7**		**115.4**	**73.1**	
Debt service-to-exports ratio (percent)	**2.1**	**1.6**	**2.3**			**3.2**	**4.8**	**4.4**	**4.0**	**4.5**	**5.3**		**7.3**	**9.1**	
PPG debt service-to-exports ratio (percent)	**2.1**	**1.6**	**2.3**			**3.2**	**4.8**	**4.4**	**4.0**	**4.5**	**5.3**		**7.3**	**9.1**	
PPG debt service-to-revenue ratio (percent)	**5.7**	**3.1**	**3.9**			**6.1**	**9.3**	**8.7**	**7.6**	**8.1**	**9.1**		**8.9**	**7.9**	
Total gross financing need (billions of U.S. dollars)	-0.3	0.8	-0.1			0.0	0.2	0.3	0.3	0.2	0.3		0.5	1.1	
Non-interest current account deficit that stabilizes debt ratio	7.9	28.8	-0.7			-0.9	3.4	0.3	-5.2	-4.9	-3.5		4.3	3.9	
Key macroeconomic assumptions															
Real GDP growth (percent)	0.9	13.0	0.5	7.9	10.7	5.4	7.7	10.5	8.2	3.4	3.5	6.4	3.2	4.5	3.4
GDP deflator in U.S. dollar terms (change in percent)	-16.1	6.6	23.2	12.3	12.1	-2.7	-0.3	2.4	1.0	-2.6	-1.5	-0.6	1.2	2.4	1.3
Effective interest rate (percent) [5]	1.4	1.0	2.2	1.2	0.4	3.3	2.9	2.8	2.7	2.7	2.7	2.8	2.1	1.8	2.0
Growth of exports of G&S (U.S. dollar terms, percent)	-30.9	23.0	22.9	49.8	84.1	1.2	-1.6	13.1	13.7	-7.8	-5.9	2.1	-2.7	4.5	-0.9
Growth of imports of G&S (U.S. dollar terms, percent)	-1.5	62.8	-21.4	27.6	53.5	-0.4	-0.8	3.2	-1.6	-0.8	-3.7	-0.7	1.2	4.4	2.1
Grant element of new public sector borrowing (percent)	...	...	...	...	...	16.0	15.2	14.4	15.1	16.2	15.5	15.4	18.9	20.3	19.4
Government revenue (excluding grants, percent of GDP)	16.1	23.6	26.6			23.1	20.7	20.5	22.5	21.4	20.6		18.3	17.6	18.3
Aid flows (billions of U.S. dollars) [7]	0.3	0.3	0.3			0.2	0.2	0.2	0.3	0.3	0.3		0.4	0.6	
of which: Grants	0.2	0.1	0.2			0.2	0.2	0.2	0.2	0.3	0.3		0.3	0.4	
of which: Concessional loans	0.1	0.1	0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.2	
Grant-equivalent financing (percent of GDP) [8]	...	...	...			2.3	2.1	2.0	2.0	2.2	2.2		2.0	1.8	2.1
Grant-equivalent financing (percent of external financing) [8]	...	...	...			46.7	47.0	47.7	50.0	52.2	52.6		73.3	58.4	61.9
Memorandum items:															
Nominal GDP (billions of US dollars)	7.1	8.6	10.6			10.9	11.7	13.2	14.4	14.5	14.8		17.3	29.7	
Nominal dollar GDP growth	-15.4	20.5	23.8			2.5	7.3	13.1	9.3	0.6	2.0	5.8	4.5	7.0	4.8
PV of PPG external debt (billions of US dollars)			2.5			2.8	2.9	3.0	3.2	3.3	3.4		3.6	3.8	
(PVt-PVt-1)/GDPt-1 (percent)						2.1	1.2	1.3	1.0	0.9	0.8	1.2	-0.3	0.3	0.1
Gross workers' remittances (billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (percent of GDP + remittances)	...	...	25.3			26.0	24.9	23.1	22.1	22.8	23.2		21.1	12.9	
PV of PPG external debt (percent of exports + remittances)	...	...	56.7			58.9	61.5	57.2	52.6	59.3	65.4		94.0	83.8	
Debt service of PPG external debt (percent of exports + remittances)	...	...	2.3			3.2	4.8	4.4	4.0	4.5	5.3		7.3	9.1	

Sources: Country authorities; and staff estimates and projections.

[1] Includes both public and private sector external debt.

[2] Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

[4] Assumes that PV of private sector debt is equivalent to its face value.

[5] Current-year interest payments divided by previous period debt stock.

[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7] Defined as grants, concessional loans, and debt relief.

[8] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Alternative Scenario

Table 6b. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032

(Percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	26	25	23	22	23	23	**21**	13
A. Alternative scenarios								
A1. Key variables at their historical averages in 20122–2032 [1]	26	28	30	32	34	35	**35**	26
A2. New public sector loans on less favorable terms in 20122–2032 [2]	26	26	24	24	25	26	**26**	22
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	26	27	29	28	29	29	**26**	16
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	26	37	56	53	55	55	**50**	21
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	26	25	23	22	23	24	**21**	13
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	26	27	24	23	24	24	**22**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	26	17	6	6	7	7	**6**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	26	36	33	32	33	33	**30**	18
PV of debt-to-exports ratio								
Baseline	59	62	57	53	59	65	**94**	84
A. Alternative scenarios								
A1. Key variables at their historical averages in 20122–2032 [1]	59	70	75	76	89	99	**158**	167
A2. New public sector loans on less favorable terms in 20122–2032 [2]	59	63	60	57	66	74	**118**	146
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	59	61	57	52	59	65	**94**	84
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	59	137	358	326	365	400	**573**	354
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	59	61	57	52	59	65	**94**	84
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	59	68	59	54	61	68	**97**	84
B5. Combination of B1-B4 using one-half standard deviation shocks	59	38	14	14	16	18	**27**	54
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	59	61	57	52	59	65	**94**	84
PV of debt-to-revenue ratio								
Baseline	113	120	113	98	107	113	**115**	73
A. Alternative scenarios								
A1. Key variables at their historical averages in 20122–2032 [1]	113	136	148	142	159	170	**194**	146
A2. New public sector loans on less favorable terms in 20122–2032 [2]	113	123	119	106	118	127	**145**	127
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	113	133	142	123	134	141	**145**	92
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	113	178	274	237	255	268	**273**	120
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	113	119	114	100	108	114	**117**	74
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	113	133	117	102	110	116	**119**	74
B5. Combination of B1-B4 using one-half standard deviation shocks	113	81	30	28	31	33	**35**	50
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	113	172	161	140	152	161	**165**	104

Alternative Scenario

Table 6b. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032 (continued)

(Percent)

Debt service-to-exports ratio								
Baseline	3	5	4	4	5	5	**7**	9
A. Alternative scenarios								
A1. Key variables at their historical averages in 20122–2032 [1]	3	4	4	4	4	4	**5**	7
A2. New public sector loans on less favorable terms in 20122–2032 [2]	3	5	4	4	4	5	**9**	12
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	3	5	4	4	5	5	**7**	9
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	3	7	13	15	16	19	**48**	49
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	3	5	4	4	5	5	**7**	9
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	3	5	5	4	5	5	**8**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	3	3	4	**1**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	3	5	4	4	5	5	**7**	9
Debt service-to-revenue ratio								
Baseline	6	9	9	8	8	9	**9**	8
A. Alternative scenarios								
A1. Key variables at their historical averages in 20122–2032 [1]	6	8	8	7	6	7	**6**	6
A2. New public sector loans on less favorable terms in 20122–2032 [2]	6	9	8	7	8	9	**11**	10
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013–2014	6	10	11	10	10	11	**11**	10
B2. Export value growth at historical average minus one standard deviation in 2013–2014 [3]	6	9	10	11	11	13	**23**	17
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2013–2014	6	9	9	8	8	9	**9**	8
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2013–2014 [4]	6	9	9	8	8	9	**9**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	6	9	8	6	6	7	**2**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	6	13	12	11	12	13	**13**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	10	10	10	10	10	10	**10**	10

Sources: Country authorities; and staff estimates and projections.

[1] Variables include real GDP growth, growth of GDP deflator (U.S. dollar terms), non-interest current account in percent of GDP, and non-debt-creating flows.

[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

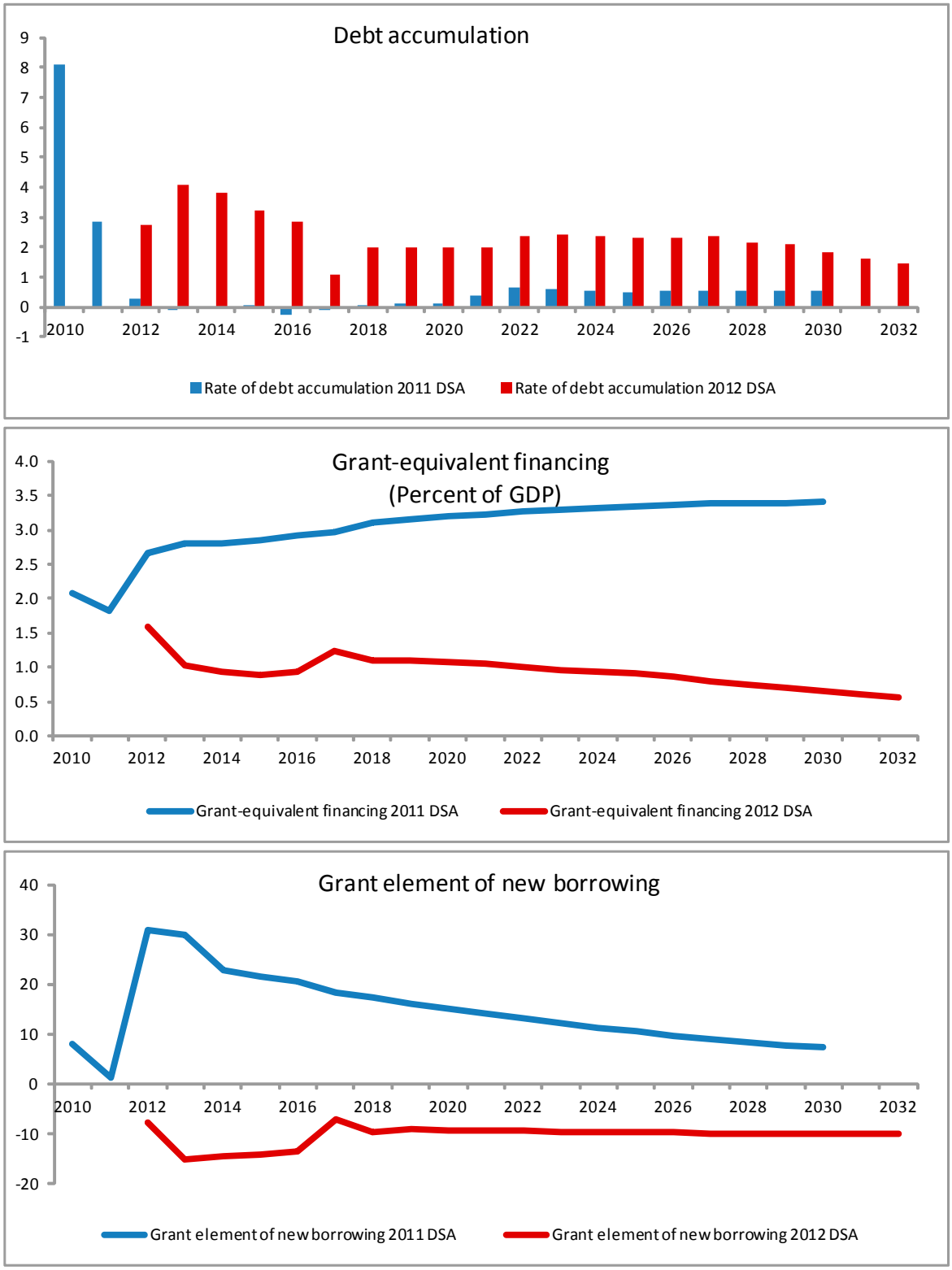
[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

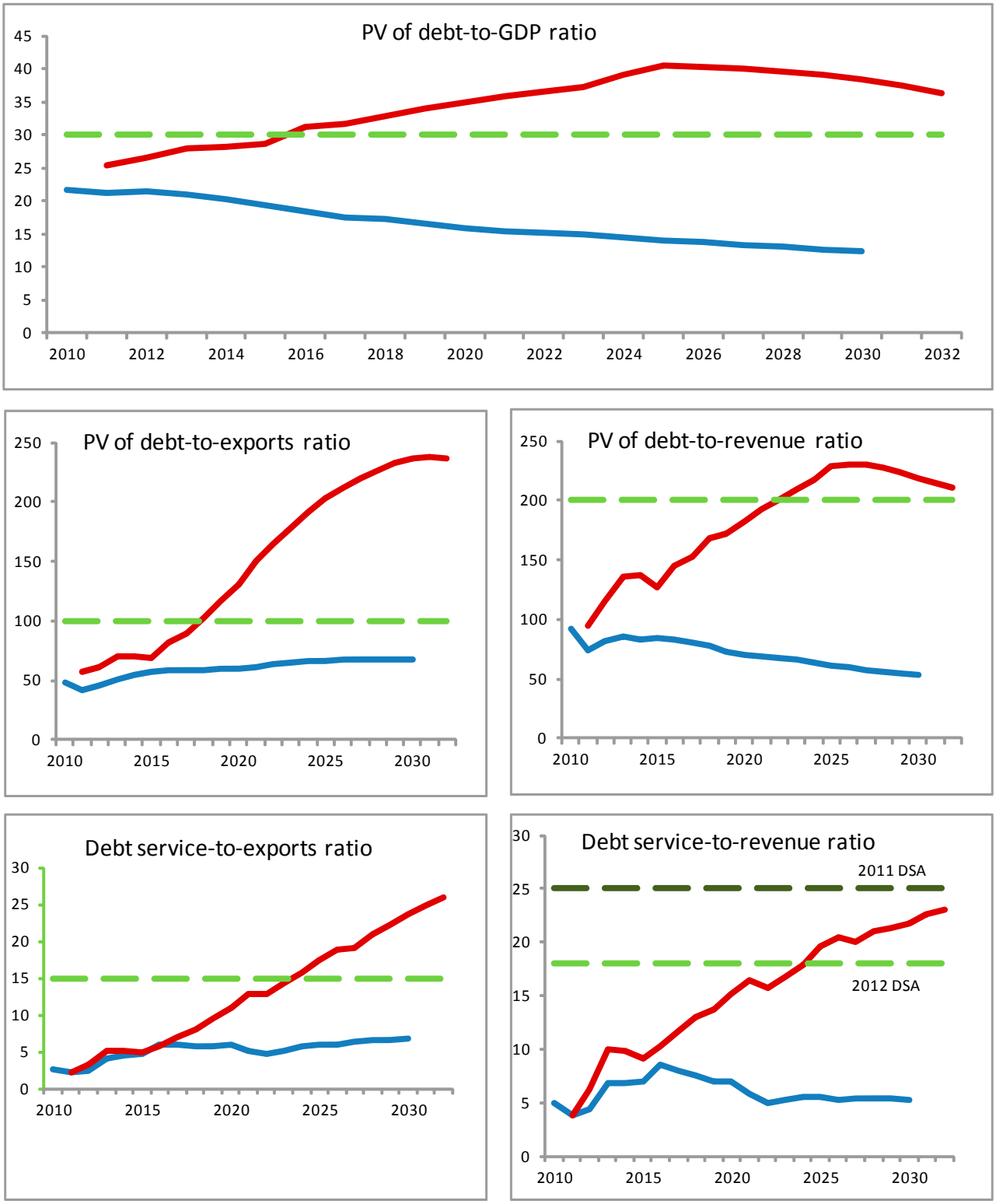
[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 5. Chad: Changes Relative to the 2011 Baseline



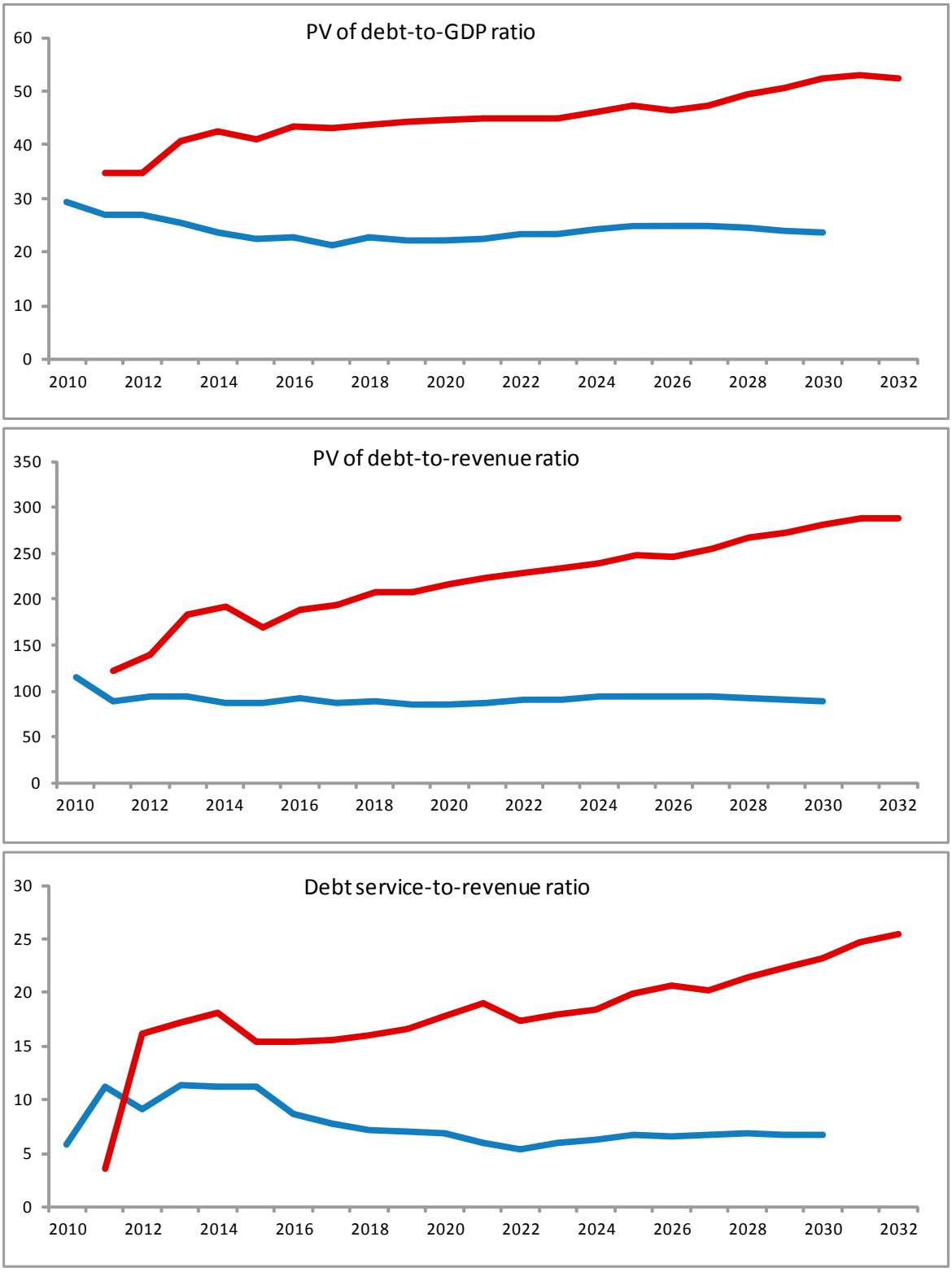
Source: Country authorities; and staff estimates and projections.

Figure 6. Chad: Changes Relative to the 2011 Baseline



Source: Country authorities; and staff estimates and projections.

Figure 7. Chad: Changes Relative to the 2011 Baseline



Source: Country authorities; and staff estimates and projections.